Standard Operating Procedure

SPR Cone Receipt and Inspection

Document ID:	SOP-SPR-001
Revision:	4
Effective Date:	2023-03-01
Supersedes:	SOP-SPR-001 Rev 3 (2021-09-15)
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Review Date:	2025-03-01

1. Purpose

This procedure defines the requirements for receiving, inspecting, and releasing SPR (Solid Phase Receptacle) cones used in the manufacture of NEXUS immunoassay kits. The procedure ensures that all SPR cones meet quality specifications before being introduced into production.

2. Scope

Applicable to all SPR cone deliveries from approved suppliers for all NEXUS product lines, including but not limited to NEXUS CK MB (30421), NEXUS B2M (30420), NEXUS Troponin (30450), and NEXUS D-Dimer (30460). This SOP applies to the Incoming Quality Control (IQC) team at the Marcy-l'Etoile production facility.

3. References

- SPEC-SPR-CKMB: SPR Cone Specification for NEXUS CK MB
- SPEC-SPR-B2M: SPR Cone Specification for NEXUS B2M
- SOP-QA-005: Incoming Material Sampling Plan
- SOP-QA-010: Non-Conformity Reporting
- FORM-SPR-001: SPR Incoming Inspection Record

4. Equipment Required

Equipment ID	Description	Use
EQ-IQC-001	UV-Vis Spectrophotometer	Antibody density measurement
EQ-IQC-002	Binding Capacity Tester	Functional binding assay
EQ-IQC-003	Sterility Test Cabinet	Sterility testing
N/A	Calibrated balance (0.1 mg)	Weight verification
N/A	Visual inspection station	Physical appearance check

5. Procedure

5.1 Receipt Verification

5.1.1 Verify that the delivery documentation (packing list, Certificate of Analysis) matches the purchase order.
5.1.2 Verify shipping conditions: temperature indicator within 15-25 C range, no visible damage to outer packaging.
5.1.3 Record receipt in ERP system and assign incoming inspection number.
5.1.4 Place material in quarantine storage pending inspection completion.

5.2 Visual Inspection

5.2.1 Sample per SOP-QA-005 (AQL 1.0, Level II).
5.2.2 Inspect for: physical damage, discolouration, foreign particles, correct labelling, correct cap colour (product-specific).
5.2.3 Record results on FORM-SPR-001.

5.3 Analytical Testing

5.3.1 **Antibody Density Measurement**
Measure antibody density using UV-Vis spectrophotometry at 280 nm.
Record result on FORM-SPR-001.

WARNING: Acceptance criterion: Antibody density must be within 2.0 - 2.5 mg/mL. Values outside this range shall be rejected per SOP-QA-010.

5.3.2 **Binding Capacity Test**
Perform functional binding assay per method AM-SPR-003.

WARNING: Acceptance criterion: Binding capacity must be >= 95%. Values below 95% shall be rejected per SOP-QA-010.

5.3.3 **Sterility Test**
Perform sterility test per pharmacopoeia method. Incubation period: 14 days.

6. Acceptance Criteria Summary

Parameter	Specification	Unit
Visual Appearance	No defects, correct colour	Pass / Fail
Antibody Density	2.0 - 2.5 mg/mL	mg/mL
Binding Capacity	>= 95%	%
Sterility	No growth after 14 days	Pass / Fail
Endotoxin	< 0.5 EU/mL	EU/mL
Weight (per cone)	1.8 - 2.2 g	g

7. Disposition

7.1 If all acceptance criteria are met: release material in ERP, move to approved storage area, update lot tracking log.
7.2 If any criterion fails: reject the lot, raise NCR per SOP-QA-010, notify supplier, initiate return process.
7.3 Borderline results (within specification but near limits): flag for Quality Engineering review. Document assessment on FORM-SPR-001 with justification for accept/reject decision.

8. Revision History

Rev	Date	Author	Changes
1	2019-01-15	J. Martin	Initial release
2	2020-06-01	J. Martin	Added binding capacity test
3	2021-09-15	J. Martin	Updated sampling plan reference
4	2023-03-01	J. Martin	Added borderline result handling (section 7.3); updated antibody density spec

9. Approval

Role	Name	Signature	Date
Author	J. Martin		
Reviewer	L. Renard		
Quality Director	P. Dubois		